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**COURSE NAME: SOFTWARE
ENGINEERING**

ASSESSMENT TOOL - 01

Project Title

AI-Powered Personalized Nutrition and Health Recommendation Platform

1. Problem Analysis

The healthcare industry is shifting toward personalized healthcare solutions, but many people still receive general diet recommendations that do not meet their individual health needs. Factors such as age, gender, weight, lifestyle, medical history, allergies, physical activity, and fitness goals differ from person to person. Traditional nutrition plans often fail to consider these differences, leading to poor eating habits, obesity, diabetes, heart diseases, and other lifestyle-related illnesses.

Healthcare professionals also spend significant time analysing patient data and preparing personalized meal plans. Manual monitoring of nutrition intake is time-consuming and prone to errors. Therefore, there is a need for an AI-powered platform that can automatically analyse health information, generate personalized nutrition plans, continuously monitor dietary habits, and provide intelligent health recommendations.

2. Stakeholder Analysis

Stakeholder	Roles	Benefits
Users/Patients	Enter health information, receive meal plans, monitor nutrition	Personalized diet recommendations and improved health
Dieticians/Nutritionists	Monitor patients, update nutrition plans	Reduced workload and accurate recommendations
Doctors	Review patient health reports	Better patient care and treatment decisions
Healthcare Organizations	Manage nutrition services	Improved service quality and efficiency
Wearable Device Providers	Share health data (steps, heart rate, calories)	Better integration with healthcare platforms
System Administrator	Manage users, data security, system maintenance	Smooth operation of the platform

3. Current System Analysis

Existing System

- Nutrition advice is mostly generic.
- Meal plans are manually prepared.
- Limited monitoring of daily food intake.
- Separate applications are used for diet tracking and fitness tracking.
- Health data from wearable devices is not fully integrated.
- Users must manually calculate calories and nutrition values.

Limitations

- No personalized recommendations.
- Time-consuming for healthcare professionals.
- High possibility of human errors.
- Lack of continuous health monitoring.
- Poor user engagement.
- Difficulty in tracking long-term nutrition progress.

4. Proposed Software Solution

The proposed system is an AI-Powered Personalized Nutrition and Health Recommendation Platform that uses Artificial Intelligence, Machine Learning, cloud computing, and wearable device integration to provide customized nutrition guidance.

Main Features

- User registration and health profile creation.
- Collection of personal health information.
- AI-based nutrition requirement analysis.
- Personalized meal plan generation.
- Daily calorie and nutrition tracking.
- Food intake monitoring.
- Integration with wearable devices.
- Health progress dashboard.
- Personalized health recommendations.

- Notifications and reminders for meals and water intake.
- Secure cloud storage for health records.

Technologies

- **Frontend:** HTML, CSS, JavaScript, React
- **Backend:** Python (Flask/Django) or Node.js
- **Database:** MySQL/MongoDB
- **AI/ML:** Python, Scikit-learn, TensorFlow
- **Cloud:** AWS/Firebase
- **Wearable Integration:** Fitbit API, Google Fit, Apple Health API

5. SDLC/Framework Justification

Selected SDLC Model: Agile Model

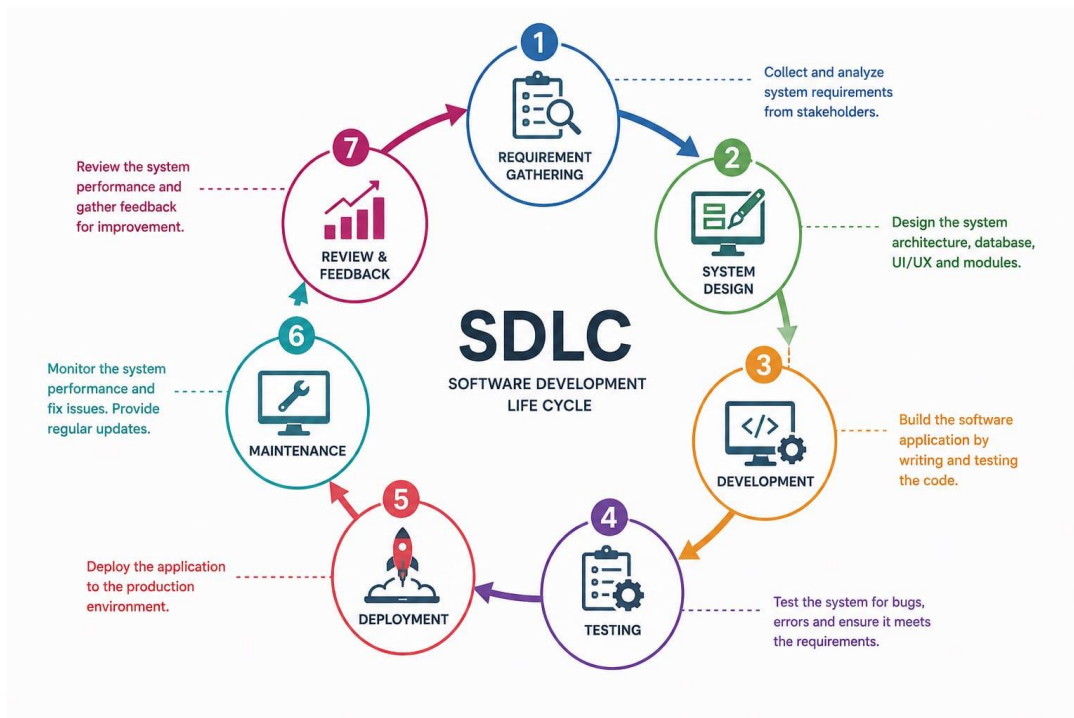
The Agile SDLC model is the most suitable because healthcare applications require continuous improvements based on user feedback and changing medical requirements.

Reasons for Choosing Agile

- Supports frequent updates.
- Easy integration of new AI models.
- Faster development using iterative cycles.
- Continuous testing improves software quality.
- User feedback can be incorporated quickly.
- Flexible to changing healthcare requirements.

Agile Phases

1. Requirement Gathering
2. Planning
3. System Design
4. Development
5. Testing
6. Deployment
7. Maintenance and Continuous Improvement



6. Expected Outcomes and Limitations

Expected Outcomes

- Personalized nutrition guidance.
- Improved dietary habits.
- Better health management.
- Reduced lifestyle-related diseases.
- Continuous nutrition monitoring.
- Increased user engagement.
- Enhanced healthcare accessibility.
- Improved overall well-being.
- Better decision-making using AI insights.
- Accurate health progress tracking.

Limitations

- AI recommendations depend on accurate user data.
- Requires internet connectivity for cloud services.
- Wearable device compatibility may vary.
- Privacy and security of health data must be ensured.

- AI cannot replace professional medical advice.
- Initial development and maintenance costs are high.
- Machine learning models require regular updates for better accuracy.
- Some users may not consistently update their dietary information.

Conclusion

The AI-Powered Personalized Nutrition and Health Recommendation Platform provide an intelligent solution for personalized healthcare by combining Artificial Intelligence, wearable technology, and cloud computing. It helps users maintain healthy eating habits, monitor nutrition intake, and improve overall well-being through customized recommendations. The Agile SDLC model ensures continuous enhancement of the platform, making it scalable, user-friendly, and adaptable to future healthcare innovations.